Oyewole Oluwaferanmi Emmanuel

SENIOR FULL-STACK & SPACE SYSTEMS ENGINEER • ASTRODYNAMICS & C2 • AUTONOMOUS EDGE AI

Location: Lagos, Nigeria | **Email:** bigferry24@gmail.com | **Phone:** +234 906 417 0048 | **Work Authorization:** COE & Global Mobility Ready
Portfolio: freshstart2066-create.github.io | **GitHub:** github.com/freshstart2066-create | **LinkedIn:** linkedin.com/in/oyewole-emmanuel

PROFESSIONAL ENGINEERING PROFILE

Senior Full-Stack & Space Systems Engineer with 4+ years of production experience architecting real-time satellite Command & Control (C2) platforms, high-throughput 3D WebGL / geospatial visualization suites, autonomous edge AI runtimes, and low-latency distributed telemetry pipelines. Combines a rigorous quantitative **B.Sc. in Statistics** with orbital astrodynamics and hardware-aware systems engineering, delivering end-to-end mission software from SGP4/SDP4 state vector propagation and WGS84 3D sensor swath projections to automated ground station pass downlinking schedulers and on-device telemetry triage.

TECHNICAL EXPERTISE & SYSTEMS TOOLKIT

Space & Astrodynamics	SGP4/SDP4 Orbit Propagation, NORAD TLE Ingestion, WGS84 Geodetic Transforms (ECEF <-> LLA <-> ECI), Nadir & SAR Swath Geometry, Ground Station Horizon Elevation Masks (AOS/LOS), Keplerian State Vectors, Constellation Tasking
Frontend & Spatial 3D	React 18, Next.js, TypeScript, Three.js, WebGL 2.0, Vite, Tailwind CSS, Canvas API, Interactive Ground Tracks, Viewport Optimization (Sustained 60 FPS WebGL Rendering)
Backend & Telemetry	FastAPI, Python (AsyncIO), Go, Node.js, WebSockets, gRPC, Server-Sent Events (SSE), RESTful APIs, Redis, PostgreSQL, TimescaleDB, Docker, Linux/Unix
AI & Autonomous Edge	Ollama, Local Quantized LLMs (Qwen 2.5, Llama 3.2), Multi-Agent DAG Orchestration, Structured JSON Tool Calling, Edge Telemetry Anomaly Detection, Streaming Inference
Systems & Reliability	Hardware-Aware Resource Tuning, Memory Buffer Management, Git, CI/CD Automated Pipelines, High-DPI Occlusion Culling, Delay-Tolerant Networking (DTN)

FLAGSHIP SPACE & AUTONOMOUS SYSTEMS

Project SERAPH • Autonomous Satellite C2 & Mission Planning Suite [Live Demo]

Three.js • WebGL 2.0 • React 18 • TypeScript • FastAPI • SGP4

- **Architected an end-to-end satellite Command & Control (C2) and mission planning suite** fusing real-time orbital astrodynamics with interactive 3D WGS84 planetary visualization running at sustained 60 FPS without external commercial licensing.
- **Implemented high-precision SGP4/SDP4 propagation engine** in backend & client, continuously computing satellite state vectors (velocity, position), predictive multi-orbit ground tracks, and ascending/descending node crossings from live NORAD TLE feeds.
- **Engineered Nadir reconnaissance and SAR sensor geometry simulation**, projecting dynamic ground footprint cones, off-nadir look angles, and imaging swaths onto the oblate WGS84 ellipsoid model for target acquisition feasibility.
- **Built automated Ground Station Pass Planning & Downlinking Scheduler**, calculating Azimuth, Elevation, Range (AER), and horizon elevation masks across global antenna networks (Tokyo, Svalbard, Chile, Hawaii) with AOS/LOS countdown timers.
- **Developed low-latency telemetry streaming architecture** over WebSockets and gRPC, broadcasting sub-second position coordinates, vehicle health status, and mission tasking access windows to operator consoles.

Project ELYRA • Multi-Provider Local Edge AI Runtime & Tactical Assistant [Live Demo]

React Native (Expo) • FastAPI • Python AsyncIO • Ollama • Quantized LLMs

- **Engineered an adaptive edge inference runtime** intelligently routing operational telemetry queries between local quantized models (Qwen 2.5, Llama 3.2 on Ollama) and cloud APIs based on latency, network availability, and zero-cloud air-gap constraints.
- **Designed sliding-window token management and memory buffer allocation**, preventing out-of-memory crashes and maintaining coherent multi-turn context on resource-constrained Android ARM64 edge hardware.
- **Implemented autonomous agentic tool execution**, structured JSON schema parsing, local hardware task sequencing, and sub-50ms streaming diagnostics over WebSockets and Server-Sent Events (SSE).

Project CHERUB • Planetary Aerostat Sounding Mesh [Planet Portal]

Distributed Systems • Aerostat Mesh • High-Temp SiC Avionics

- **Coordinated planetary aerostat probe mesh architecture** for the corrosive 50–65 km Venusian cloud decks, featuring autonomous buoyancy station-keeping control and delay-tolerant space packet networking.
- **Architected resilient fail-safe telemetry beacons** engineered for extreme pressure-temperature thermal envelopes utilizing Silicon Carbide (SiC) logic specifications and distributed inter-probe relay synchronization.

CloudPulse & RaftConsensus • Distributed Systems Observability & Consensus [GitHub]

React 18 • TypeScript • SVG Telemetry • Raft State Machine

- **Built real-time telemetry observability platform** with dynamic SVG sparkline charts, global edge node mesh tracking across 6 worldwide PoPs, and automated incident triage with 1-click remediation triggers.
- **Engineered full Raft consensus protocol state visualizer** in TypeScript, modeling Leader Election, Log Replication, Heartbeat Timers, and interactive network chaos partition injection for mission-critical cluster resilience.

PROFESSIONAL ENGINEERING EXPERIENCE

Independent Software & Systems Engineering Consultant — Jan 2024 – Present

Senior Full-Stack & Systems Developer • Lagos, Nigeria / International Remote

- **Architect and deliver high-performance full-stack applications, geospatial dashboards, and real-time microservices** for international engineering and enterprise clients.
- **Implement responsive, mission-critical front-end architectures** using React, Next.js, and TypeScript, improving page-load speeds by up to 35% and ensuring 99.8% crash-free sessions across heterogeneous hardware profiles.
- **Design resilient asynchronous backend services with FastAPI, Go, and Node.js**, integrating WebSockets and SSE for streaming high-frequency telemetry, sensor feeds, and autonomous agentic workflows.
- **Orchestrate containerized deployments using Docker and Linux/Unix environments**, enforcing rigorous CI/CD automation, comprehensive test coverage, and deterministic execution.

Dantech — Jan 2022 – Dec 2023

Contract Software Engineer • Remote / Hybrid

- **Developed modular REST and WebSocket APIs using Python and Node.js** for multi-tenant internal applications supporting 1,000+ daily active users with sub-100ms response targets.
- **Refactored legacy frontend codebases into modular, reusable React component systems**, reducing bundle sizes and improving Core Web Vitals (LCP, FID) by 30%.
- **Optimized PostgreSQL database schemas, indexing strategies, and complex query execution plans**, slashing report generation turnaround times by 45% and eliminating query deadlocks under peak transaction volume.
- **Integrated automated CI/CD validation pipelines** enforcing rigorous linting, type safety checks, and integration test suites prior to production container deployment.

SpaceEDU & Decentralized Protocol Research — 2022 – 2023

Lead Systems Engineer & Simulator Architect • Open-Source / Independent Research

- **Engineered interactive 3D orbital mechanics educational simulators (SpaceEDU)** utilizing Three.js and numerical integration to demonstrate Hohmann transfer orbits, Lagrange points, and gravitational perturbations.
- **Researched and built decentralized cryptographic identity and state verification prototypes**, establishing verifiable audit logs for distributed sensor and hardware telemetry payloads.

EDUCATION & QUANTITATIVE RIGOR

Ekiti State University (EKSU) • Ado Ekiti, Nigeria — Graduated 2024

Bachelor of Science (B.Sc.) in Statistics

- **Rigorous Statistical Training:** Comprehensive coursework in Probability Theory, Statistical Computing, Operations Research, Mathematical Modeling, Regression Analysis, and Stochastic Processes.
- **Mathematical Application to Space Systems:** Principles directly applied to orbital state vector propagation, Monte Carlo conjunction collision probability (Pc) screening, sensor swath spherical trigonometry, and hardware performance benchmarking.

SYSTEMS ARCHITECTURE PROFICIENCIES & METHODOLOGIES

- **High-Reliability Architecture:** Event-driven telemetry processing, asynchronous worker queues, fail-safe degradation strategies, and deterministic state synchronization across distributed nodes.
- **Performance & Spatial Optimization:** Frustum culling, level-of-detail (LOD) geometry streaming, WebGL buffer pooling, memory-efficient vector arrays, and sub-50ms UI response rendering.

GLOBAL MOBILITY, WORK AUTHORIZATION & LANGUAGES

- **Work Visa & COE Sponsorship Ready:** Fully eligible for international Work Visa / Certificate of Eligibility (COE) sponsorship. All academic degree certificates, verified transcripts, and government credentials prepared for rapid filing.
- **Global Relocation:** Prepared for immediate relocation to Japan (Tokyo / Tsukuba), Europe, North America, or hybrid international positions. Holds valid International Driving Permit (IDP).
- **Languages:** English (Native / Bilingual Fluency); Japanese (Committed Beginner / Actively studying conversation & technical terms).